Tips on Writing Papers in LaTeX

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Abstract

We list some best practices and tips for writing technical papers in LaTeX.

Index Terms

Tips, suggestions, best practices.

I. General

Please see the tex file for actual implementation of the examples in this note. You will not learn anything by just reading the pdf file. This tex file includes some of the more commonly used packages and settings. It also illustrates the use of some packages, e.g., the enumitem package allows customization of the item listing. For details, read the documentation of the packages for different options. discrete set $\{1, 2, \ldots, n\}$.

1) Windows tex distribution: MiKTeX.
2) There are many good LaTeX tutorials online. A good starting point is http://tobi.oetiker.ch/lshort/lshort.pdf.
3) There are many tex editors you can use. I highly recommend VSCode (with git integration), which you can use for other coding purposes as well. More traditional latex-centric editors are TeXstudio or TeXnicCenter. You can also make use of online latex platforms like Overleaf. In this case, please ensure that you always have a local updated backup copy. The server can crash right before a paper deadline leaving you with no access (it has happened before and will happen again!).
4) Use IEEEtran draftcls as in this tex file for drafts (editing stage). See also http://www.michaelshell.org/tex/ieeetran/ for faqs.

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5) I recommend the use of pdflatex instead of latex. This means your figures should be in pdf or png format instead of eps. The reason is that typical dvi viewers are not very stable and responsive. However, if you insist on using latex, that is also fine. I recommend using the latexmk toolchain ("latexmk -synctex=1 -interaction=nonstopmode -file-line-error -pdf") to automatically perform the necessary multiple pdflatex runs.
6) After running pdflatex or latexmk, **please look through the warnings and resolve them**. These are very helpful to catch errors like defining same labels multiple times.
7) I have provided a preamble.tex that you can include into your tex file using the \input command (see the beginning of this tex file). This preamble.tex file includes all the commonly used packages, declarations and aliases. **Look through the preamble.tex file first before you define your own newcommands to avoid duplication. The most updated preamble.tex file is always available on https://gitlab.com/wptay/latex-tips.** I will always use the most updated preamble.tex file, so there is no need for you to include a copy in your workspace (you can keep a global copy somewhere and configure your Latex to point to it or include a path in latexmkrc). If you find some useful things to add to preamble.tex, please let me know.
8) A typical working tree looks like that in Fig. 1.
9) You can use this tex file as a starting template when writing your papers.
10) Upload all references you have used in your paper to Google Drive, except for books. If you have referenced a paper, you should have at least read its abstract and the relevant parts that you have referenced.
11) Please ensure that you embed all your fonts into the pdf before you submit to a journal or conference.

II. Labels

(a) Use meaningful labels for your sections, equations, inequalities, figures, tables, and others. For example, use something like "sect:Labels" to label your section so that you know that this label is for a section. Later on in your text, you can then easily refer to it via something like "Section II".

Example 1. *Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque.*

```
work directory
├─ figures
│  ├─ abc.pdf
│  └─ xyz.png
├─ drafts (NOT required if using git)
│  ├─ papernameV1.tex
│  ├─ papernameV1.pdf
│  ├─ papernameV2.tex
│  ├─ papernameV2.pdf
│  ├─ responseR1.tex
│  └─ responseR1.pdf
├─ conferencename (conference version)
│  ├─ latexmkrc (for BIBINPUTS and TEXINPUTS)
│  ├─ papernameconf.tex
│  └─ papernameconf.pdf
├─ bib (if you have many bib files)
│  ├─ file1.bib
│  └─ file2.bib
├─ latexmkrc (for BIBINPUTS and TEXINPUTS)
├─ papername.tex
├─ papername.pdf
├─ supplementary.tex (if applicable)
└─ response.tex (if applicable)
```

Fig. 1. Typical working tree.

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla *ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.*

(b) Tip: items can also be labeled! Example is this reference (b).

(c) Do not over label! For equations, tables, figures and most "objects", we usually label only those that we need to refer to in another part of the text. In some cases, you may also label important equations although you may not refer to these in any part of your text. This makes it easier for readers to refer to your equations. However, do this judiciously.

(d) You should try to use the cleveref package to reference labels. It is smart enough to know what you are referencing, e.g., Section I, Sections I to III, and (1) all use the same cref command. Otherwise, you have put a tilde like "Section I" to avoid line breaks immediately after "Section".

(e) Refer to Algorithm 1.

Algorithm 1 Minibatch stochastic gradient algorithm

Initialize $\boldsymbol{\theta}, \boldsymbol{\phi}$.

Set $\mathbf{y} = f(\boldsymbol{\theta})\boldsymbol{\phi}$.

repeat

  Draw a mini-batch data points $\{(\mathbf{x}_{j_i}, \mathbf{s}_{j_i}) : i = 1, \ldots, M\}$ from the training dataset and M independent and identically distributed (i.i.d.) noise samples $\{\boldsymbol{\xi}_i : i = 1, \ldots, M\}$ from $\mathcal{N}(\mathbf{0}, \mathbf{I})$. ▷ This is a comment.

  Estimate $\mathbf{y}$ by using the M data points $\{(\mathbf{x}_{j_i}, \mathbf{s}_{j_i}) : i = 1, \ldots, M\}$.

  Update $\boldsymbol{\theta}, \boldsymbol{\phi}$ by applying stochastic gradient descent to optimize (1).

until $\boldsymbol{\theta}$ converges

III. Equations

1) There are many ways to write equations, but I find that you really only need to use either the align or align* environments provided by amsmath. The environment align will have

equation numbers at every line and align* has no equation numbers. See the tips on labeling to decide when to use align and align*. Examples are:

$$
\begin{aligned}
X &= \sum_{i=1}^{n} y_i \qquad (1)\\
&= z\\
&= w, \qquad (2)
\end{aligned}
$$

and

$$
\begin{aligned}
X &= \int_a^b f(x)\,\mathrm{d}x\\
&= z\\
&= w.
\end{aligned}
$$

Another example is

$$
\begin{aligned}
\Lambda_{\mathrm{SGLR}}(k,t) &= \max\left\{\Lambda(k,t), \Lambda_n(k,t)\right\}, \qquad (3)\\
\tau_{\mathrm{SGLR}}(b) &= \min\{\tau(b), \tau_n(b)\}, \qquad (4)\\
\tau_{\mathrm{W-SGLR}}(b) &= \min\{\widetilde{\tau}(b), \widetilde{\tau_n}(b)\}, \qquad (5)\\
\Phi_i &= \tau_{\mathrm{W-SGLR}}(b) + i^2, \text{ for } i = 1, \ldots, N,\\
\widetilde{\mathbf{C}} &= \text{one\_hot}\left(\widetilde{\mathbf{L}}\right), \qquad (6)\\
d &= D(p_X \,\|\, p_Z).
\end{aligned}
$$

Note when punctuation marks like "," and "." are required in the equations. Observe also how text is written in math environments.

2) Please put & before the binary operators like $=$, $\leq$ and $\geq$. You can see why by comparing the following example with that above. Do you see that the spacing behind $=$ is now missing?

$$
\begin{aligned}
X =& \int_a^b f(x)\,\mathrm{d}x\\
=& z\\
=& w.
\end{aligned}
$$

3) Do not write something like

$$\sum_{i=1}^{n} y_i$$

$$= z$$

$$= w,$$

and therefore, we have

$$x + y = w.$$

It should be written as follows:

$$\sum_{i=1}^{n} y_i$$

$$= z$$

$$= w,$$

and therefore, we have

$$x + y = w.$$

Alternatively, use the intertext command:

$$\sum_{i=1}^{n} y_i$$

$$= z$$

$$= w,$$

and therefore, we have

$$x + y = w.$$

4) When to use \text and \mathrm? From tex.stackexchange:

\ text { divides } and \mathrm{divides} might give the same result , but they are conceptually different (and can actually be printed in different ways, depending on the math fonts used) . Spaces in the argument of \mathrm are ignored, for example. Moreover, \ text honors the font of the surrounding environment: it will print in italics in the statement of a theorem.

You should use abc if abc is part of a math symbol but abc if abc is text like in a conditional clause in a math environment.

5) Please refrain from using the "$$ ... $$" or "\[... \]" environment for equation displays. It makes the latex hard for me to read and makes it difficult to add labeling later on. Tip: If you find it a hassle to type the begin and end tags, set up some keyboard shortcuts! You should have keyboard shortcuts for the commonly used environments and commands to improve your writing efficiency. A useful tool is AutoHotkey.
6) An interesting package is breqn, which provides special equation environments that can perform line breaks *automatically*. This is very useful if you need to produce both single- and double-column versions of the same paper. Do check it out! As there are still some bugs in this package, I recommend that you use the breqn environments only for those few long equations, while sticking to align and align* for the rest.

IV. New Commands

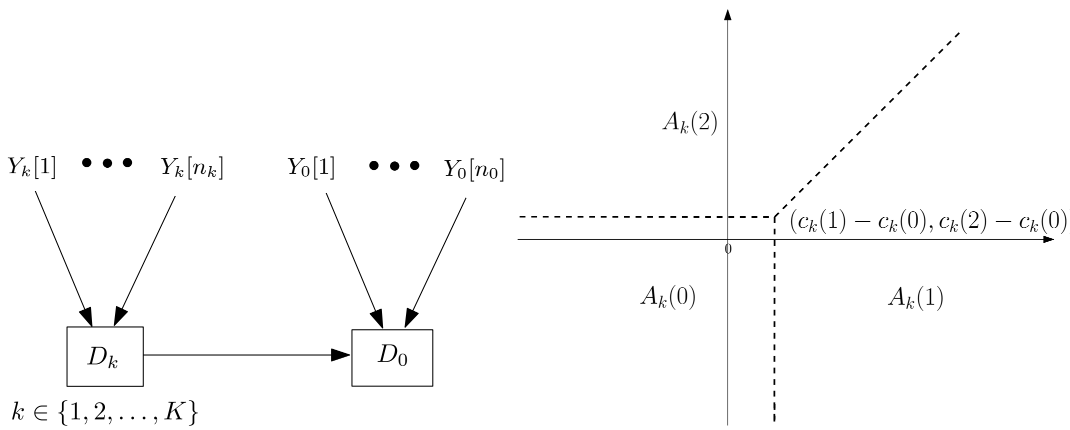
1) If you are using a group of symbols very frequently, define a new command for it. It is like writing a computer program. The new command can even take in an argument.
2) Example of a simple new command is $\mathbb{R}$. See the provided preamble.tex file for more examples!
3) Example of a more complex new command is $\mathbb{E}_0[f(X)]$. **As we use $\mathbb{E}$ and $\mathbb{P}$ very frequently, please figure out how I have defined these commands in the preamble.tex and use them appropriately.** Examples are:

$$
\begin{aligned}
f(\boldsymbol{\theta}) &= \mathbb{E}^{\beta}_{X\sim p_{\boldsymbol{\theta}}}\left[\frac{f(X)}{g(X,Y)}\,\middle|\,Y\right], \\
f(\boldsymbol{\theta}) &= \mathbb{E}_{X\sim p_{\boldsymbol{\theta}}}\left[\frac{f(X)}{g(X,Y)}\,\middle|\,Y\right], \qquad \text{(this will be deprecated, use \textbackslash given)} \\
\mathbb{P}(A \,|\, B) &= \mathbb{P}\left(\left\{\omega\in\Omega \,:\, \frac{1}{2}X(\omega) > b\right\}\,\middle|\, \limsup_{n\to\infty} X_n(\omega) = a\right).
\end{aligned}
$$

V. Graphics

1) Put your figures into a subfolder called "Figures".
2) You can set the graphics path in tex using "\graphicspath" (this has been done in the preamble.tex file). Then LaTeX will look for the right figure in the folder you have provided in the graphics path. See Fig. 2 for an example.

3) In Fig. 2, we use the subfigure environment to create Figs. 2a and 2b.
4) I highly recommend Ipe (http://ipe.otfried.org/) for preparing your graphics. The advantage is that you can directly input latex symbols into your graphics using Ipe. Graphics can be saved directly in pdf. For example, you can open tandem.pdf in the Figures folder using the Ipe program and edit that! There is also an Ipelet that allows you to convert Ipe to tikz if you prefer the latter. If you use another graphics program, the font may not be the same as your latex document. In that case, you may need to use psfrag to replace your labels.
5) Please use "!htb" or "!htbp" as options in your figures environment.
6) Use the subcaption package for the subfigure environment. Do not use the subfig package as it does not work with the hyperref package. See example in Fig. 2.



(a) Network topology　　(b) Decision regions

Fig. 2. A tandem network and its decision regions.

VI. Paper Editing Process

1) After you have written your paper, put it aside for a day and revise it with a fresh mind in order to improve it. **This includes checking the bibliography for errors.** See Section VIII.
2) Please send me your paper for editing only after you have gone over it **several times** to correct any errors and improve it. You should send the paper to me only after you feel fully satisfied with the version that you have. Typically, this will be the fourth or fifth version.
3) Share your tex files with me through *private repositories* on Gitlab, GitHub, Gitee or BitBucket and add me as a member or collaborator. Git takes care of version control so that we do not have to resort to manual version control. Alternatively, you can share an

Overleaf project with me. I prefer the git repository as it allows me to use my own editor (with many more useful features and keyboard shortcuts!) than the web interface editor of Overleaf. The free version of Overleaf also does not come with version control beyond the 24-hour window.

4) I will highlight my major edits in blue and red, where the latter color usually indicates something that requires your action or special attention. Please go through all my edits to check for errors (never assume that I am always correct; I do make errors too!). If you are satisfied with an edit, remove the highlight and return the file to me, with your own edits highlighted in blue. We will iterate this process until the paper reaches a satisfactory stage.
5) Please note that sometimes **NOT** all my edits are highlighted. Those highlighted are for your attention and checking. Some are for you to compare with your previous version so that you can learn from the change. Do not manually transfer the changes to your previous tex file. Use the tex file I send you for your next iteration.

VII. Responding to Paper Reviews

1) You can use the response.tex as a starting template. I have defined some useful environments for use there in your responses to reviewers' comments.
2) Check out the use of the packages catchfilebetweentags and xr (or xr-hyper if the hyperref package is used). These packages make life much easier when you need to cross-reference a label or citation from your manuscript tex file in your response file. It also allows you to quote a chunk of tex directly from your manuscript tex file. How cool is that?
3) In the response.tex file, replace manuscript-filename with your .tex filename (without the ".tex"). Add tags in your .tex file to reproduce the same chunk between the tags in your response. E.g., if you add the tags in your tex file:

```
%<*tag:NP>
Therefore, we have shown that $P \ne NP$.
%</tag:NP>
```

and use "\revision{tag:NP}" in your response.tex, the sentence between the two tags will be quoted in your response. This ensures that whatever changes you make in the tex file are reflected in the response. Examples given below are used in response.tex.

Example 2. *Consider the subset of vertices V_0 of an unweighted directed graph G. Suppose that we consider the adjacency matrix $\mathbf{A}_G$ as the graph shift operator for G. The adjacency matrix $\mathbf{A}_{H_0}$ of the subgraph H_0 shown is a submatrix consisting of the even-indexed rows and columns of $\mathbf{A}_G^2$. Any filter shift invariant with respect to (w.r.t.) $\mathbf{A}_{H_0}$ can be viewed as a polynomial of $\mathbf{A}_G^2$ composed with a projection to the vertices of H_0.*

Theorem 1. *For any $\Gamma \in L^2(X \times [n])$, $\Gamma * = \mathbb{E}_{\mu_X}[\Gamma_x *]$ is a bounded linear operator.*

Proof: See Section SII in the supplementary material. ■

The following is another theorem.

Theorem 2. *Suppose $T \subset \mathbb{R}$ is a finite or a bounded interval. If almost surely every $x \in X$ has no repeated eigenvalues and has uniformly bounded operator norm, then every convolution filter is a bi-polynomial filter.*

VIII. Bibliography

The statement "\def\loadbibentry" allows the bib entries to be printed as below. Remove or comment this out in your paper.

1) Use bibtex to keep track of your references.
2) Use a meaningful system for your bibtex keys. For example, a common practice is to use the first 3 letters of the first 3 authors' last name followed by the year the paper is published, e.g., "LenTayQui:J16".
3) Use the IEEEabrv.bib file. Read the file to understand how it works. I have also provided a StringDefinitions.bib that contains some of the conferences relevant to our research areas.
4) Programs like JabRef can help you to organize your bib library.
5) In the following, I list examples of correct or preferred reference citations as well as bad citations in the IEEEtran bibliographystyle. Please refer to refs.bib to see the bib entries. Can you tell the difference between the correct citation and each of the bad citations?
   (i) Example of correct journal citation:
      - M. Leng, W. P. Tay, F. Quitin, C. Cheng, S. G. Razul, and C. M. S. See, "Anchor-aided joint localization and synchronization using SOOP: Theory and experiments," *IEEE Trans. Wireless Commun.*, vol. 15, no. 11, pp. 7670 – 7685, Nov. 2016

   (ii) Examples of bad journal citation:

- M. Leng, W. P. Tay, F. Quitin, C. Cheng, S. G. Razul, and C. M. S. See, “Anchor-aided joint localization and synchronization using soop: Theory and experiments,” *IEEE Trans. Wireless Commun.*, vol. 15, no. 11, pp. 7670 – 7685, Nov. 2016
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- M. Leng, W. P. Tay, F. Quitin, C. Cheng, S. G. Razul, and C. M. S. See, “Anchor-aided joint localization and synchronization using SOOP: Theory and experiments,” *IEEE Transactions on Wireless Communications*, vol. 15, no. 11, pp. 7670 – 7685, Nov. 2016 [Note: If you do not want to use standard abbreviations in the journal name, you have to be consistent and do this for ALL journals and conferences.]
- M. Leng, W. P. Tay, F. Quitin, C. Cheng, S. G. Razul, and C. M. S. See, “Anchor-aided joint localization and synchronization using SOOP: Theory and experiments,” *IEEE trans. wireless commun.*, vol. 15, no. 11, pp. 7670 – 7685, Nov. 2016

(iii) Examples of correct conference citation:

- M. Sun and W. P. Tay, “Privacy-preserving nonparametric decentralized detection,” in *Proc. IEEE Int. Conf. Acoustics, Speech, and Signal Processing*, Shanghai, China, Mar. 2016
- M. Leng, F. Quitin, C. Cheng, W. P. Tay, S. G. Razul, and C. M. S. See, “Joint navigation and synchronization using SOOP in GPS-denied environments: Algorithm and empirical study,” in *Proc. Sensor Signal Processing for Defence Conf.*, Edinburgh, UK, Sep. 2015

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- M. Leng, F. Quitin, C. Cheng, W. P. Tay, S. G. Razul, and C. M. S. See, “Joint navigation and synchronization using SOOP in GPS-denied environments: Algorithm and empirical study,” in *Sensor Signal Processing for Defence Conf.*, Edinburgh, UK, Sep. 2015
- M. Leng, F. Quitin, C. Cheng, W. P. Tay, S. G. Razul, and C. M. S. See, “Joint navigation and synchronization using SOOP in GPS-denied environments: Algorithm and empirical study,” in *Proc. 5th Sensor Signal Processing for Defence Conf.*, Edinburgh, UK, Sep. 2015 [Note: If you want to keep the “5th”, you have to be consistent and do this for ALL conferences.]
- M. Leng, F. Quitin, C. Cheng, W. P. Tay, S. G. Razul, and C. M. S. See, “Joint navigation and synchronization using SOOP in GPS-denied environments: Algorithm and empirical study,” in *Proc. Sensor Signal Processing for Defence Conf.*, Sep. 2015

(v) Example of correct book citation:

- D. Stroock, *Probability Theory: An Analytic View*. Cambridge, UK: Cambridge University Press, 1993

(vi) Examples of bad book citation:

- D. Stroock, *Probability Theory: an Analytic View*. Cambridge, UK: Cambridge University Press, 1993
- D. Stroock, *Probability Theory: An Analytic View*. Cambridge, UK: Cambridge university press, 1993
- D. Stroock, *Probability Theory: An Analytic View*. Cambridge University Press, 1993